Your TA will lend your group a polystyrene model of a solid bead $S \subseteq \mathbb{R}^3$. **Please handle it with care and do not draw on it. Return it to your TA at the end of tutorial.**

Orientation: The z-axis passes through the centre of the bead's cylindrical hole.

T16.1 Using your model with your teammates, determine how to visualize S in cylindrical coordinates.

(a) Your TA will share the equations that define ∂S in rectangular form (x, y, z). Record them below.

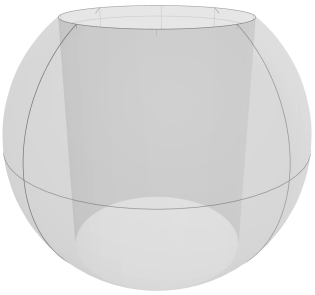
- The equation _____ corresponds to the bead's spherical outer surface.
- The equation _____ corresponds to the bead's cylindrical hole.

(b) Rewrite the equations defining ∂S in cylindrical form (r, θ, z).

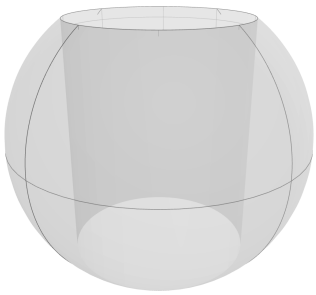
- The equation _____ corresponds to the bead's spherical outer surface.
- The equation _____ corresponds to the bead's cylindrical hole.

(c) Draw some typical r-slices, θ-slices, and z-slices of S on the diagrams below.
Include at least **two** slices of each type.

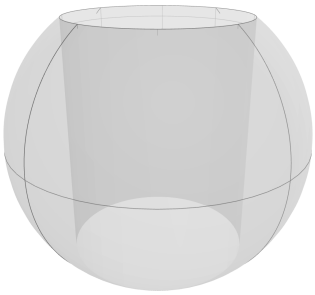
r-slices



θ-slices



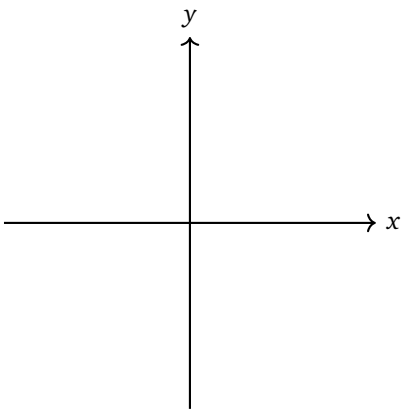
z-slices

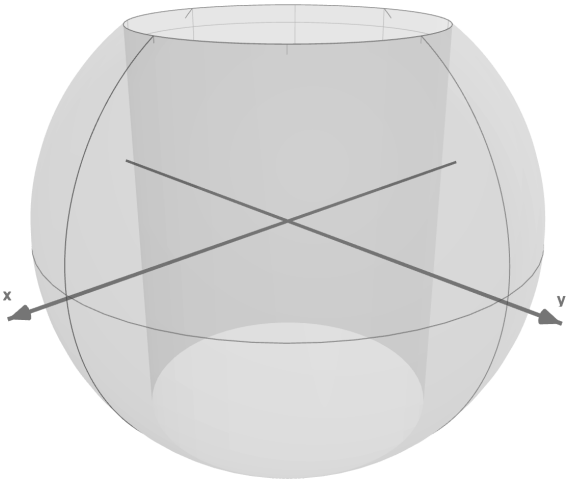


(d) For which values of $z \in \mathbb{R}$ is the z-slice of S non-empty? _____

Express the inequalities defining the z-slice of S in polar form (r, θ). _____

(e) On the left, draw the projection $P \subseteq \mathbb{R}^2$ of S onto the (x, y)-plane. Fix a point $(x, y) \in P$ in the projection.
On the right, draw the vertical wire inside S that projects onto (x, y).





(f) Express the projection $P \subseteq \mathbb{R}^2$ in polar form. _____

(g) Which order(s) of integration correspond to the "slicing method with z-slices"?
Circle the correct choice(s).

$dr\,d\theta\,dz$ $dr\,dz\,d\theta$ $d\theta\,dr\,dz$ $d\theta\,dz\,dr$ $dz\,dr\,d\theta$ $dz\,d\theta\,dr$

(h) Which order(s) of integration correspond to the "projection onto the (x, y)-plane"?
Circle the correct choice(s).

$dr\,d\theta\,dz$ $dr\,dz\,d\theta$ $d\theta\,dr\,dz$ $d\theta\,dz\,dr$ $dz\,dr\,d\theta$ $dz\,d\theta\,dr$

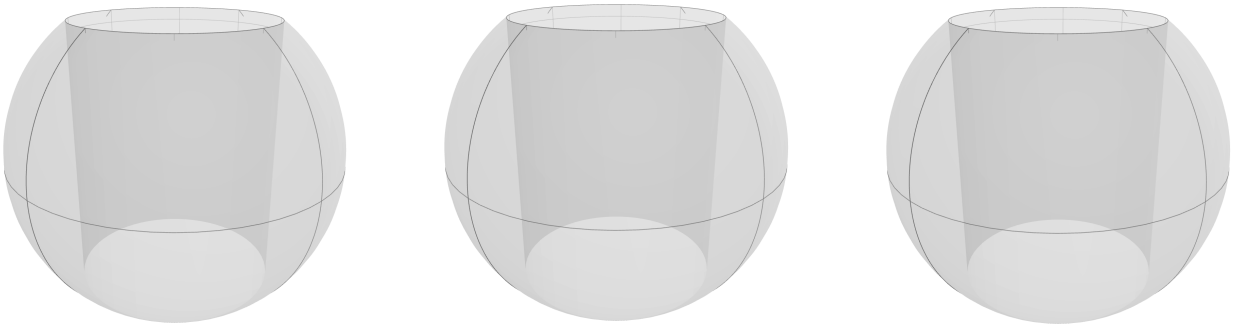
T16.2 Continue using your model with your teammates to visualize S in spherical coordinates.

(a) Rewrite the equations defining ∂S in spherical coordinates (ρ, θ, ϕ).

- The equation _____ corresponds to the bead's spherical outer surface.
- The equation _____ corresponds to the bead's cylindrical hole.

(b) Draw some typical ρ-slices, θ-slices, and ϕ-slices of S on the diagrams below.
Include at least **two** slices of each type.

ρ-slices θ-slices ϕ-slices



(c) Which order(s) of integration correspond to the "slicing method with ρ-slices"? Circle the correct choice(s).

$d\rho\,d\theta\,d\phi$ $d\rho\,d\phi\,d\theta$ $d\theta\,d\rho\,d\phi$ $d\theta\,d\phi\,d\rho$ $d\phi\,d\rho\,d\theta$ $d\phi\,d\theta\,d\rho$

(d) Which order(s) of integration correspond to the "slicing method with θ-slices"? Circle the correct choice(s).

$d\rho\,d\theta\,d\phi$ $d\rho\,d\phi\,d\theta$ $d\theta\,d\rho\,d\phi$ $d\theta\,d\phi\,d\rho$ $d\phi\,d\rho\,d\theta$ $d\phi\,d\theta\,d\rho$

(e) Which order(s) of integration correspond to the "slicing method with ϕ-slices"? Circle the correct choice(s).

$d\rho\,d\theta\,d\phi$ $d\rho\,d\phi\,d\theta$ $d\theta\,d\rho\,d\phi$ $d\theta\,d\phi\,d\rho$ $d\phi\,d\rho\,d\theta$ $d\phi\,d\theta\,d\rho$

(f) For which values of $\phi \in \mathbb{R}$ is the ϕ-slice of S non-empty? _____

Express the ϕ-slice of S in spherical form. _____

T16.3 Finally, you can assemble all of your previous progress to express the volume of S in both coordinate systems.
(Your TA will share a Math3D demo, which you can use in addition to your model to help visualize.)

(a) Express the volume of S as an iterated integral in **cylindrical** coordinates $dr\,d\theta\,dz$. **Do not evaluate yet.**

(b) Express the volume of S as an iterated integral in **spherical** coordinates $d\rho\,d\theta\,d\phi$. **Do not evaluate yet.**

(c) Calculate the integrals from parts (a) and (b). Should you obtain the same values? Briefly explain.
(You will probably need more space than provided. Use WolframAlpha if you're short on time in tutorial.)